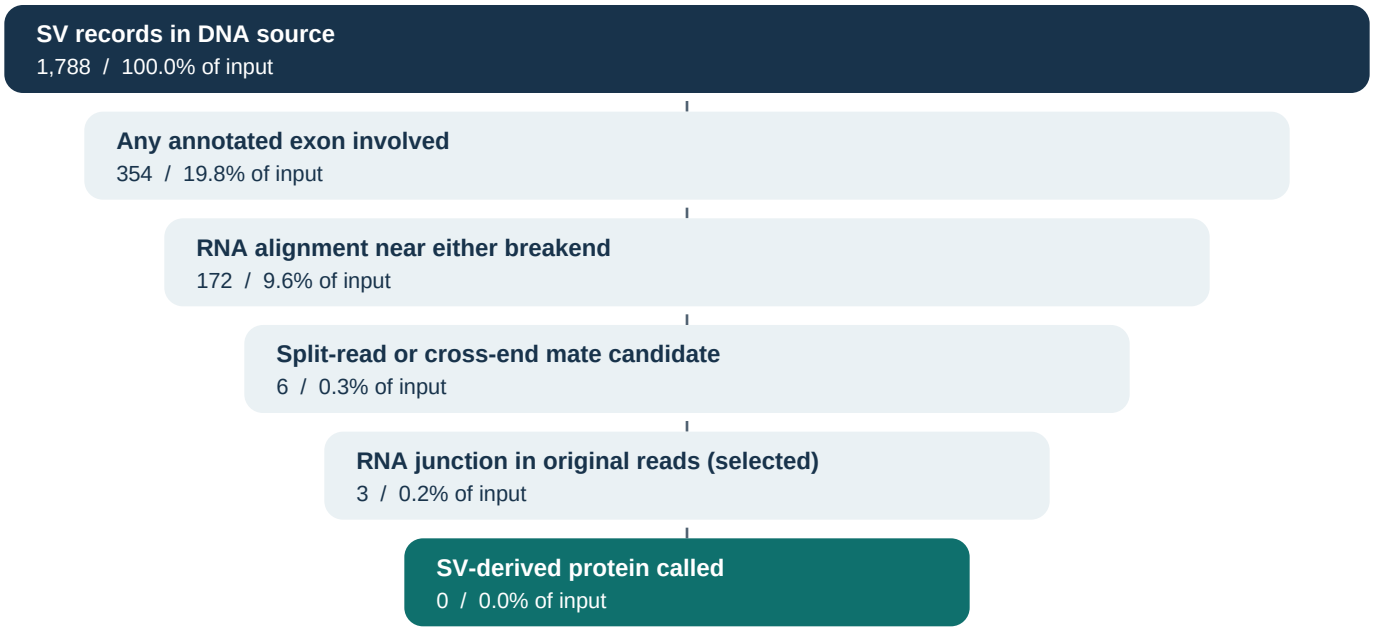
HCC1395: somatic mutation and RNA evidence

Whole source-call audit | matched-normal DNA baseline | original tumor RNA | pinned reference ORFs. Counts below describe the specified public call sets and RNA run, not every biological mutation or a validated full-length isoform.

Small variants: DNA to RNA and local protein

Boxes show subset order; widths are schematic, not scaled to count.

Structural variants: DNA to RNA breakpoint evidence



Boxes show subset order; widths are schematic, not scaled to count.

Each funnel uses its own DNA source-record denominator. Breakend-window coverage is locus coverage, not evidence for the SV. A split-read or cross-end mate candidate links nominated windows but does not resolve junction bases, orientation, splice path or a translated product. The junction stage is a targeted original-read replay of selected events; it is a lower bound for the full source set.

At a glance

Evidence layer	Count	Interpretation
Ingested small variants	41,072	39,447 SNVs; 1,625 indels
Ingested SV source records	1,788	DNA breakpoint records/events, counted in their source sets
SVs involving an annotated exon	354	Span for DEL/DUP/INV; nominated breakpoint or insertion flanks otherwise
Any annotated exon involved	1,728	At least one changed base or insertion flank overlaps an Ensembl 87 exon
Ensembl 87 coding effect	491	Any annotated protein-coding transcript, reference model
orfdata ncORF coding-block overlap	17	Candidate reference ORF overlap; distinct from mutant translation
Original RNA ALT-observed calls	343	2,681 ALT reads; 2,681 ALT fragments
Calls with local protein contexts	22	119 ranked contexts; full-length isoforms unresolved

Coding annotation and its denominator

Of 41,072 DNA small-variant records, 491 change the coding sequence of at least one Ensembl 87 transcript. The pinned orfdata 0.34.1 GRCh38 catalog (7,264 reference ncORF models; release 2026-10-11.1) has exact coding-block overlaps at 17 calls. The groups overlap at 3 calls, so the union is 505. These are annotation opportunities. The catalog does not assess a cell-line-specific mutant ORF, RNA expression, or peptide evidence.

Reference annotation	All DNA calls	With original RNA ALT
Ensembl coding effect	491	135
orfdata coding-block overlap	17	5
Union of either category	505	see per-call table

SNVs query their changed reference base. Deletions query removed reference bases; insertions use the VCF anchor as a candidate overlap. The per-call table preserves that rule. A coding-block hit does not prove the inserted sequence is in frame.

Any annotated exon is involved in 1,728 calls (4.2% of the small-variant source set). This includes UTR and noncoding-transcript exons. Deletions query every removed reference base, including exon crossings with intronic endpoints; insertions query both neighboring reference bases as candidate exon involvement.

Original RNA and local protein evidence

RNA assay: SRR23456709, ONT direct RNA. Each nominated small variant has a status from this run. An ALT read is a sequence observation; it is not a full-length mutant isoform.

RNA allele status	Small-variant records
alternate observed	343
reference only	239
reference and other only	53
other allele only	19
alignment without allele call	7,300
no alignment overlap	33,118

The 343 ALT-observed calls contribute 2,681 allele-read and 2,681 fragment observations. Isovar resolves at least one read-compatible **local** protein context for 22 calls (119 ranked rows). The remaining 321 ALT-observed calls have no local translation under the stated method. No-overlap and uncallable statuses cannot establish absence of RNA expression.

OSTC F9L and PRDX5 F157L have original-read local contexts. An independent proteomic study observed the mutant residues, not these entire contexts. The bounded ONT cDNA fixture is excluded from whole-run counts.

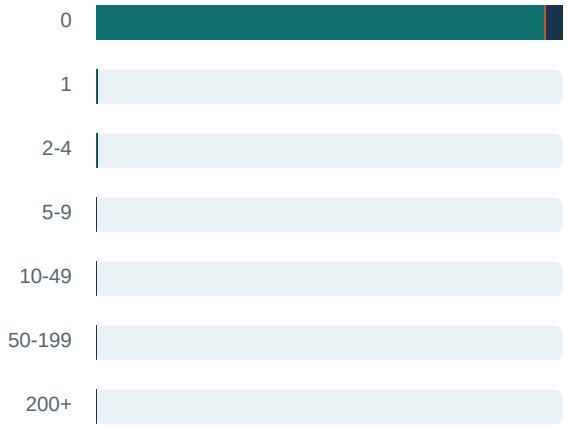
Structural-variant RNA screen

Breakend screen outcome	Source records
No breakend-window alignment	1,233
One end covered; no split candidate	149
Both ends covered; no split candidate	400
Split candidate; path unresolved	6

Six Talsania records have same-read split-alignment candidates (17 record-level observations). Published POU2F1-NME7 has one candidate read. Junction bases, orientation, splice choice and ORF remain unresolved; no fusion protein is called.

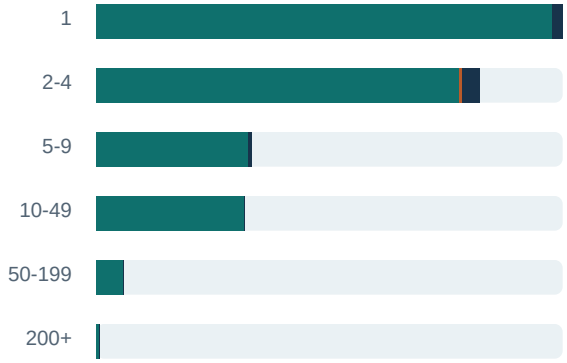
RNA alternate-read support by variant class

ALT reads per variant: all records



	SNV	insertion	deletion
0	39,114	156	1,459
1	132	0	3
2-4	105	1	5
5-9	44	0	1
10-49	43	0	0
50-199	8	0	0
200+	1	0	0

ALT reads per variant: positive support only

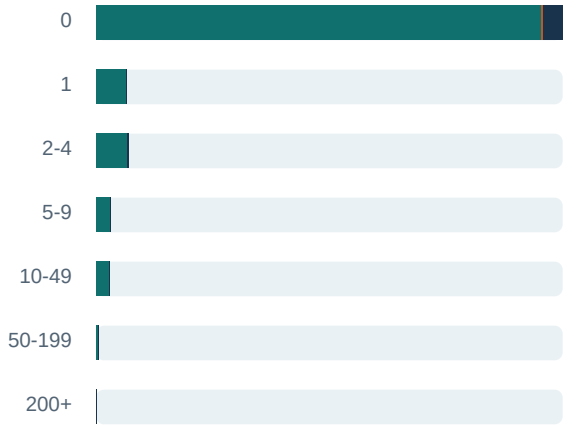


	SNV	insertion	deletion
1	132	0	3
2-4	105	1	5
5-9	44	0	1
10-49	43	0	0
50-199	8	0	0
200+	1	0	0

The positive-support panel excludes zero-ALT records. Bars stack SNV, insertion, deletion and other-replacement record counts. Matching colored columns print each class count directly. These are variant counts, not read counts.

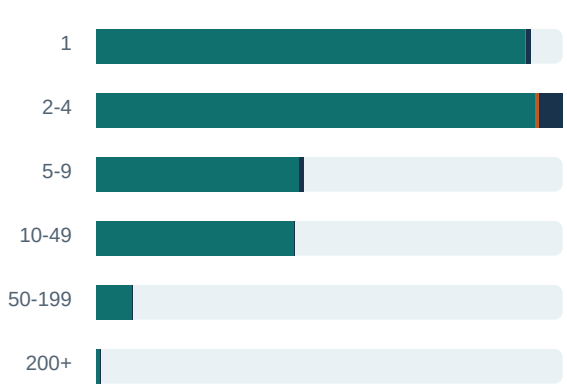
RNA ALT reads: exon-involved small variants

ALT reads per exon-involved variant: all records



	SNV	insertion	deletion
0	1,371	4	61
1	93	0	1
2-4	95	1	5
5-9	44	0	1
10-49	43	0	0
50-199	8	0	0
200+	1	0	0

ALT reads per exon-involved variant: positive support only



	SNV	insertion	deletion
1	93	0	1
2-4	95	1	5
5-9	44	0	1
10-49	43	0	0
50-199	8	0	0
200+	1	0	0

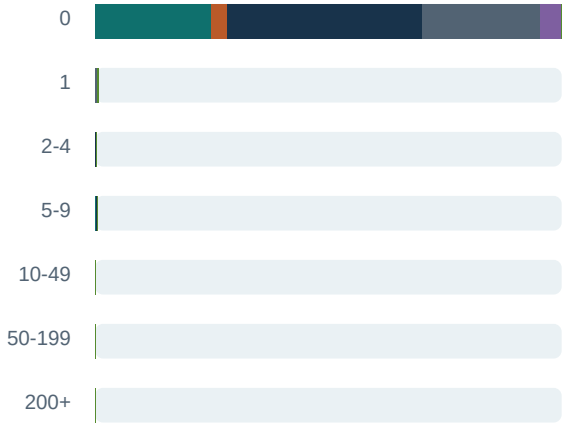
The input is the exon-involved subset of the small-variant source calls. The first panel retains zero-ALT calls; the second conditions on positive ALT reads. Insertion flanks are candidate exon involvement. SVs have a separate plot because a split alignment is not an ALT-spanning read.

Exon-involved nominated SVs: RNA breakpoint screen

354 of 1,788 nominated SV source records involve an Ensembl 87 exon. Intrachromosomal DEL/DUP/INV records query their full nominated span; INS queries insertion flanks; TRA/BND query nominated breakpoints. 0 records have insufficient breakpoint coordinates to rule out exon involvement and are excluded from this plot.

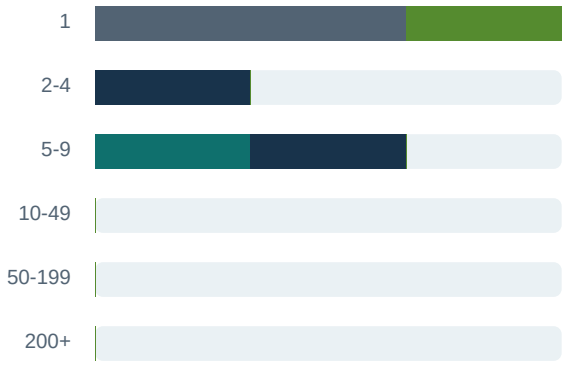
RNA screen	Exonic SV records	Candidate reads or pairs
Same-read split alignments	6	17
Cross-end primary mates	0	0

RNA link candidates per exon-involved SV: all records



	DEL	INS	DUP	INV	TRA	BND
0	87	12	145	88	16	0
1	0	0	0	2	0	1
2-4	0	0	1	0	0	0
5-9	1	0	1	0	0	0
10-49	0	0	0	0	0	0
50-199	0	0	0	0	0	0
200+	0	0	0	0	0	0

RNA link candidates per exon-involved SV: positive support only



	DEL	INS	DUP	INV	TRA	BND
1	0	0	0	2	0	1
2-4	0	0	1	0	0	0
5-9	1	0	1	0	0	0
10-49	0	0	0	0	0	0
50-199	0	0	0	0	0	0
200+	0	0	0	0	0	0

Bars count SV source records by reported class, binned on split-read plus cross-end mate candidate counts. Same-read supplementary alignments and opposite primary mates are different kinds of screening evidence; neither validates a junction sequence or translated product. The screen uses 500-base windows and mapping quality at least 20. The COLO829 benchmark call sets remain separate in the underlying table.

Original-read replay: junction versus RNA path

This selected-event replay uses oriented DNA adjacencies and actual RNA alignment records, including soft-clipped bases and low-MAPQ pieces. Direct junction fragments count reads crossing the same RNA join, even when their wider transcript sequences differ. MAPQ 20 is a sensitivity subset, not a gate on whether the junction sequence was observed. Isovar extends these joins through overlapping original read sequences; one read need not span the assembled context. Context length is a bounded assembly result, not an isoform-support count.

Nominated DNA record	Class	RNA join, all MAPQ	Join, MAPQ 20 subset	Longest context (bp)
chr13:52,688,329-60,966,095	DEL	4	3	2430
chr21:42,975,947-43,035,721	DUP	2	2	2225
chr9:83,969,153-84,015,113	DUP	7	7	2198

The chr9 duplication has seven direct junction fragments; three event paths survive assembly and the search reaches its ten-path cap. The chr13 deletion has four direct fragments, including one lost at MAPQ 20; chr21 has two. Overlapping reads extend each candidate context; their local branch support still needs evaluation. The three chr1 screen rows reuse one complex read and were not oriented here.

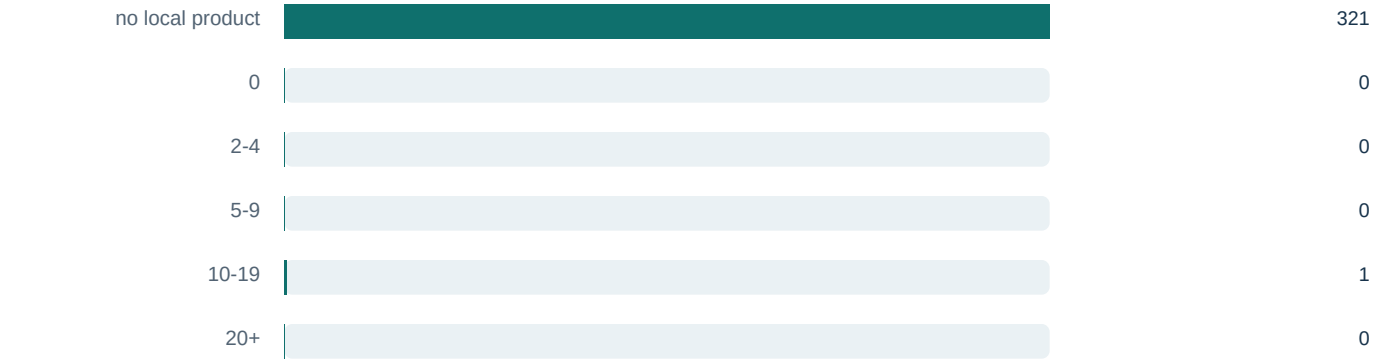
The HCC1395 replay BAM is a source-derived 309-record regional fixture; COLO829 uses the 129,534-record selected-site BAM. Both were extracted after aligning the complete original RNA run. No transcript models were supplied in this geometry audit, so frame and SV-derived protein status remain unresolved. Other splicing and DNA-to-RNA causation have not been excluded.

Isovar-marked amino-acid span

For each RNA ALT-observed call, count amino-acid positions in the mutation interval of its top-ranked local protein context. This is not an independently verified count of changed residues: synonymous DNA calls can produce a reference-identical amino acid window. Calls with no local context are shown explicitly. The second panel omits the one-residue category to expose the longer tail; its denominator is smaller. A zero-length interval can represent a deletion junction. These windows do not determine a complete frameshift tail or a full-length isoform.

Marked AA positions: all ALT-observed calls

Marked AA positions: excluding one-AA calls



Mutation classes and resolved outcomes

Category	DNA source	RNA ALT	Local context / product
DNA-predicted frameshift	9	1	See per-call local contexts
Reference ncORF coding-block overlap	17	5	Mutated ncORF not reconstructed
SV source record	1,788	Breakend screen only	0 resolved products

Frameshifts are Varcode top-effect predictions from DNA, not a count of RNA-confirmed full-length frameshift proteins. SV split/mate candidates do not count as productive until an RNA junction and protein path resolve.

Highest original-RNA ALT support with local contexts

Source call	Type	Support / self screen	Top local sequence
chr12:6867653:G>C	SNV	236 ALT reads	KQNLGSSSALSTRPRCRPTP
COSM32759	SNV	193 ALT reads	VRLLADPTGALGKETDLLLD
COSM32783	SNV	166 ALT reads	DKLKEAETRAQFAERTVAKL

Least reference-self-like local contexts

Source call	Type	Support / self screen	Top local sequence
chr8:123220221:A>AGGGCTAC	insertion	14/14 novel	HPEVKCVGSVARSPDCLGDCILRRI
chr12:6867653:G>C	SNV	9/9 novel	KQNLGSSSALSTRPRCRPTP
COSM32759	SNV	9/9 novel	VRLLADPTGALGKETDLLLD

The score is the fraction of distinct mutation-overlapping 9-mers absent from the pinned Ensembl 87 reference protein FASTA. It is an exact-match screen of these local windows, not patient-specific self, HLA binding, presentation or immunogenicity.

Highest supported non-SNV local products

Source call	Type	Support / self screen	Top local sequence
chr8:123220221:A>AGGGCTAC	insertion	3 ALT reads	HPEVKCVGSVARSPDCLGDCILRRI

The tables rank source records, not independent molecules. A displayed local context may represent only part of a translated protein and can be reference-identical for a synonymous allele.

Reference and mutant protein windows

The orange reference and teal mutant residues differ at the marked boundary; dark residues are shared flanks. Each reference window comes from the named Ensembl 87 transcript. The mutant is Isovar's top-ranked local context supported by tumor RNA reads. For frameshifts, downstream columns are anchored at the change, not a homology alignment or a complete protein. A dash means the reference protein ends before the mutant window.

COSM32759 | SNV | 193 ALT reads | ENST00000265462 from amino acid 147

```
REF  VRLLADPTGAFGKETDLLLD
MUT  VRLLADPTGALGKETDLLLD
```

COSM32783 | SNV | 166 ALT reads | ENST00000300933 from amino acid 194

```
REF  DKLKEAETRAEFAERTVAKL
MUT  DKLKEAETRAQFAERTVAKL
```

Sources, processing and limits

Input	Pinned scope
Matched-normal small variants	SEQC2 v1.2.1 SNV + indel, GRCh38.d1.vd1; HCC1395BL normal
SV benchmark	Talsania et al. 2022 Table S4, HCC1395 vs HCC1395BL
Original RNA	PRJNA934933 / SRR23456709; 1,049,452 ONT direct RNA reads
Alignment	minimap2 2.26 -ax splice -uf -k14; NCBI GRCh38 no-alt

Reference model and interpretation

Small-variant coding effects use Varcode with Ensembl release 87 on GRCh38. The human orfdata 0.34.1 catalog uses Ensembl 101 GRCh38 reference ORFs (7,264 models, dataset 2026-10-11.1). An orfdata coding-block overlap is a reference annotation candidate; this analysis did not translate mutated ncORFs from the RNA reads. Genomic transcript overlap is not RNA-compatible isoform support. Original-read allele counts and local protein contexts are separate observations.

SHA-256 input pins

SEQC2 SNV VCF gzip

`0c49f7e1519fc7860b773f79bdcf07453fe4def8303707f34542471d92494c9d`
SEQC2 indel VCF gzip

`db7dc8be2ec04bab830ff1411146c7d538c217ff684e874b85d98a3dbbda91cf`
Talsania Table S4 xlsx

`11e8d6393b83f0e0b7fdb0b13a57376663fc9662116d78c89a0d5fa530df096b`
SRR23456709 SRA archive

`d69db8e48bbb20665c3e03a1fa35742be818a0127994e015cb3a560afae39ac8`
GRCh38 no-alt FASTA (uncompressed)

`9cce8b926416dd96b152deea85188495b75f7ac8d634cc723a017067be8702b7`
orfdata bundled GRCh38 catalog

`06bc0bf3f81c25a7356f71a38581ff5704c05dcb75752454bc1ca75075ab8c09`
Ensembl 87 GRCh38 protein FASTA gzip (self screen)

`d873c199d5dbf20869db3f2cbd6ac726259215d91c2782f9f1ddcad4bb117cf9`

Other antigen sources in this run

Category	Assessment
Viral RNA and viral coding paths	Not assessed: no viral-aligned RNA panel
Supplied fusion transcripts	Not assessed with direct junction reconstruction
Targeted antigen-gene isoforms	Not assessed with original-read isoform coverage
Mutated ncORF proteins	Not reconstructed; reference coding-block overlaps only

What remains unresolved

ONT raw-read errors can create other-base calls and competing local contexts. The RNA and DNA libraries come from separate studies. A 500-base breakend window and split-alignment candidate are screening evidence only. Selected junctions were resolved with original reads, but complete isoform assignment and SV protein reconstruction remain unresolved.

Machine-readable per-call annotation, allele, protein, and SV tables and complete reconstruction commands are published with the Isovar HCC1395/COLO829 reports. These PDFs summarize selected public source call sets, not a complete catalogue of every cell-line mutation.